# PIERCE COUNTY AP, WA, USA

WMO: **720388**

Lat: **47.104N** Lon: **122.287W** Elev: **164** StdP: **99.37** Time Zone: **-8.00 (NAP)** Period: **07-19** WBAN: **00469**

## Annual Heating, Humidification, and Ventilation Design Conditions

| | Coldest Month | Heating DB | | Humidification DP/MCDB and HR | | | | | | Coldest Month WS/MCDB | | | | MCWS/PCWD to 99.6% DB | | WSF | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 99.6% | | | 99% | | | 0.4% | | 1% | | | | | |
| | | 99.6% | 99% | DP | HR | MCDB | DP | HR | MCDB | WS | MCDB | WS | MCDB | MCWS | PCWD | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (1) | **12** | **-7.0** | **-4.1** | **-10.3** | **1.6** | **-3.8** | **-7.2** | **2.1** | **-2.1** | **7.2** | **9.7** | **6.0** | **9.0** | **0.0** | **10** | **0.472** | (1) |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| | Hottest Month | Hottest Month DB Range | Cooling DB/MCWB | | | | | | Evaporation WB/MCDB | | | | | | MCWS/PCWD to 0.4% DB | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 0.4% | | 1% | | 2% | | 0.4% | | 1% | | 2% | | | | |
| | | | DB | MCWB | DB | MCWB | DB | MCWB | WB | MCDB | WB | MCDB | WB | MCDB | MCWS | PCWD | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (2) | **8** | **13.4** | **31.3** | **19.3** | **29.0** | **18.5** | **27.3** | **17.8** | **20.2** | **28.5** | **19.2** | **26.9** | **18.4** | **25.4** | **1.6** | **330** | (2) |

| | Dehumidification DP/MCDB and HR | | | | | | | | | Enthalpy/MCDB | | | | | | Extreme Max WB | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 0.4% | | | 1% | | | 2% | | | 0.4% | | 1% | | 2% | | | |
| | DP | HR | MCDB | DP | HR | MCDB | DP | HR | MCDB | Enth | MCDB | Enth | MCDB | Enth | MCDB | | |
| | (a) | (b) | (c) | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (3) | **17.5** | **12.8** | **20.9** | **17.1** | **12.5** | **20.6** | **16.2** | **11.7** | **19.8** | **58.1** | **28.7** | **55.1** | **26.8** | **52.6** | **25.1** | **23.7** | (3) |

## Extreme Annual Design Conditions

| | Extreme Annual WS | | | | Extreme Annual Temperature | | | | n-Year Return Period Values of Extreme Temperature | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Mean | | Standard Deviation | | n=5 years | | n=10 years | | n=20 years | | n=50 years | | |
| | 1% | 2.5% | 5% | | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | Min | Max | |
| | (a) | (b) | (c) | | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (4) | **5.5** | **4.7** | **3.9** | DB | **-10.2** | **35.1** | **2.0** | **2.7** | **-11.7** | **37.1** | **-12.8** | **38.7** | **-13.9** | **40.2** | **-15.4** | **42.1** | (4) |
| (5) | | | | WB | **-10.4** | **21.9** | **2.0** | **0.9** | **-11.9** | **22.5** | **-13.0** | **23.1** | **-14.2** | **23.6** | **-15.6** | **24.2** | (5) |

## Monthly Climatic Design Conditions

| | | | | Annual | Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | (n) | (o) | (p) | |
| (6) | Temperatures, Degree-Days and Degree-Hours | | DBAvg | **10.9** | **4.7** | **5.3** | **7.2** | **9.4** | **13.0** | **15.6** | **18.6** | **19.0** | **15.6** | **10.5** | **7.1** | **4.0** | (6) |
| (7) | | | DBStd | **6.00** | **3.51** | **3.38** | **2.70** | **2.66** | **2.95** | **2.87** | **2.57** | **2.49** | **2.76** | **2.63** | **3.69** | **3.82** | (7) |
| (8) | | | HDD10.0 | **753** | **166** | **134** | **95** | **41** | **5** | **0** | **0** | **0** | **1** | **25** | **100** | **187** | (8) |
| (9) | | | HDD18.3 | **2822** | **423** | **366** | **346** | **267** | **166** | **93** | **27** | **22** | **89** | **242** | **338** | **443** | (9) |
| (10) | | | CDD10.0 | **1069** | **2** | **2** | **7** | **23** | **99** | **167** | **267** | **278** | **168** | **42** | **12** | **2** | (10) |
| (11) | | | CDD18.3 | **96** | **0** | **0** | **0** | **0** | **1** | **11** | **35** | **42** | **7** | **0** | **0** | **0** | (11) |
| (12) | | | CDH23.3 | **1576** | **0** | **0** | **2** | **16** | **82** | **188** | **574** | **584** | **129** | **1** | **0** | **0** | (12) |
| (13) | | | CDH26.7 | **529** | **0** | **0** | **0** | **3** | **16** | **63** | **204** | **212** | **31** | **0** | **0** | **0** | (13) |
| (14) | Wind | | WSAvg | **0.9** | **0.9** | **1.0** | **1.3** | **1.2** | **0.9** | **0.8** | **0.7** | **0.7** | **0.6** | **0.6** | **1.0** | **0.9** | (14) |
| (15) | Precipitation | | PrecAvg | **1062** | **150** | **102** | **114** | **86** | **62** | **47** | **18** | **26** | **44** | **102** | **165** | **147** | (15) |
| (16) | | | PrecMax | **1443** | **312** | **218** | **228** | **188** | **109** | **100** | **48** | **94** | **197** | **244** | **365** | **293** | (16) |
| (17) | | | PrecMin | **793** | **60** | **4** | **39** | **25** | **7** | **8** | **1** | **1** | **3** | **19** | **42** | **31** | (17) |
| (18) | | | PrecStd | **169** | **58** | **51** | **44** | **41** | **28** | **24** | **12** | **23** | **40** | **50** | **78** | **65** | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% | DB | | **14.1** | **16.1** | **21.3** | **25.9** | **28.7** | **31.4** | **34.0** | **33.7** | **30.1** | **22.0** | **18.0** | **15.0** | (19) |
| (20) | | | MCWB | | **10.7** | **10.4** | **12.7** | **15.5** | **16.2** | **19.8** | **19.4** | **19.3** | **19.0** | **15.3** | **14.2** | **11.3** | (20) |
| (21) | | 2% | DB | | **12.6** | **13.7** | **17.1** | **21.1** | **25.9** | **28.6** | **31.4** | **32.0** | **27.1** | **18.9** | **16.0** | **12.5** | (21) |
| (22) | | | MCWB | | **9.9** | **9.8** | **10.7** | **13.5** | **16.1** | **18.8** | **19.0** | **19.5** | **17.8** | **14.0** | **12.8** | **10.2** | (22) |
| (23) | | 5% | DB | | **11.3** | **12.3** | **14.0** | **17.5** | **22.8** | **25.0** | **28.9** | **28.9** | **24.0** | **17.3** | **13.9** | **11.0** | (23) |
| (24) | | | MCWB | | **9.4** | **9.2** | **9.6** | **11.1** | **15.0** | **16.7** | **18.4** | **18.8** | **16.8** | **13.0** | **11.7** | **9.5** | (24) |
| (25) | | 10% | DB | | **9.9** | **10.9** | **12.4** | **15.0** | **20.0** | **22.5** | **27.0** | **27.1** | **22.2** | **15.9** | **12.5** | **9.0** | (25) |
| (26) | | | MCWB | | **8.7** | **9.1** | **9.0** | **9.9** | **14.3** | **15.7** | **17.7** | **17.9** | **16.0** | **12.9** | **10.9** | **8.1** | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% | WB | | **12.4** | **12.6** | **13.4** | **16.1** | **18.6** | **20.7** | **21.9** | **21.1** | **20.4** | **17.1** | **15.5** | **12.7** | (27) |
| (28) | | | MCDB | | **13.6** | **14.3** | **16.1** | **24.3** | **26.4** | **29.6** | **31.7** | **30.5** | **27.4** | **18.9** | **17.7** | **13.7** | (28) |
| (29) | | 2% | WB | | **11.0** | **11.2** | **11.8** | **13.7** | **17.0** | **19.2** | **20.0** | **20.0** | **18.9** | **15.4** | **13.6** | **11.2** | (29) |
| (30) | | | MCDB | | **11.8** | **12.5** | **15.1** | **19.3** | **23.5** | **26.4** | **28.4** | **28.7** | **24.5** | **17.1** | **15.2** | **12.3** | (30) |
| (31) | | 5% | WB | | **9.7** | **10.0** | **10.7** | **12.2** | **15.7** | **17.5** | **19.0** | **19.1** | **17.8** | **14.2** | **12.3** | **9.2** | (31) |
| (32) | | | MCDB | | **10.5** | **11.2** | **13.5** | **15.7** | **21.6** | **22.8** | **26.8** | **27.0** | **21.5** | **16.0** | **13.7** | **10.0** | (32) |
| (33) | | 10% | WB | | **8.4** | **8.8** | **9.7** | **10.9** | **14.5** | **16.4** | **18.1** | **18.4** | **16.9** | **13.0** | **11.2** | **8.0** | (33) |
| (34) | | | MCDB | | **9.2** | **10.0** | **12.1** | **13.8** | **19.2** | **22.0** | **25.1** | **25.4** | **20.2** | **14.9** | **12.4** | **8.6** | (34) |
| (35) | Mean Daily Temperature Range | | MDBR | | **7.1** | **8.0** | **9.5** | **10.0** | **11.3** | **11.3** | **13.8** | **13.4** | **11.6** | **9.8** | **8.1** | **6.6** | (35) |
| (36) | | 5% DB | MCDBR | | **8.7** | **11.1** | **13.2** | **15.9** | **17.6** | **17.1** | **18.9** | **18.4** | **16.3** | **12.4** | **9.3** | **7.4** | (36) |
| (37) | | | MCWBR | | **5.9** | **7.1** | **7.6** | **8.2** | **8.6** | **7.5** | **7.2** | **7.0** | **7.7** | **7.7** | **6.4** | **5.1** | (37) |
| (38) | | 5% WB | MCDBR | | **6.9** | **8.0** | **11.0** | **13.2** | **15.0** | **15.2** | **16.3** | **16.2** | **13.2** | **9.8** | **8.3** | **6.5** | (38) |
| (39) | | | MCWBR | | **5.4** | **5.9** | **7.0** | **7.3** | **7.7** | **7.0** | **6.9** | **6.7** | **6.8** | **7.0** | **6.2** | **5.1** | (39) |
| (40) | Clear-Sky Solar Irradiance | taub | | | **0.319** | **0.315** | **0.323** | **0.358** | **0.374** | **0.367** | **0.349** | **0.356** | **0.344** | **0.327** | **0.319** | **0.328** | (40) |
| (41) | | taud | | | **2.465** | **2.500** | **2.514** | **2.398** | **2.379** | **2.424** | **2.493** | **2.480** | **2.509** | **2.561** | **2.521** | **2.373** | (41) |
| (42) | | Ebn at Noon | | | **769** | **860** | **907** | **897** | **890** | **897** | **909** | **889** | **871** | **836** | **766** | **697** | (42) |
| (43) | | Edh at Noon | | | **67** | **78** | **89** | **110** | **116** | **112** | **103** | **100** | **89** | **73** | **63** | **65** | (43) |
| (44) | All-Sky Solar Radiation | RadAvg | | | **1.05** | **1.92** | **2.87** | **4.22** | **5.30** | **5.85** | **6.53** | **5.57** | **4.09** | **2.34** | **1.24** | **0.86** | (44) |
| (45) | | RadStd | | | **0.15** | **0.26** | **0.33** | **0.38** | **0.43** | **0.55** | **0.47** | **0.38** | **0.36** | **0.26** | **0.16** | **0.12** | (45) |

## Historical Trends

| | | DBAvg | Heating | | Cooling | | | Degree-Days | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | 99% DB | 99% DP | 1% DB | 1% WB | 1% DP | HDD10.0 | HDD18.3 | CDD10.0 | CDD18.3 | |
| | | (d) | (e) | (f) | (g) | (h) | (i) | (j) | (k) | (l) | (m) | |
| (46) | Station Only | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (46) |
| (47) | Regional (57 neighbors) | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | **N/A** | (47) |

Nomenclature: See separate page